EEEE4138 Aerial Robotics

Individual Parameter Allocation

Use the parameter ID assigned to your name for the practicals and coursework. Include this ID in your report, parameter script and model. All configurations have four rotors. Values are assigned simulation benchmarks, not measured aircraft specifications.

ID	Student name	Geometry	m0 (kg)	l (m)	Nominal mass mb (kg)
S01	Mostafa Mahmoud Hassan Mahmoud Mohamed	Plus	1.31	0.230	1.965
S02	Dennis Tie Siou Ong	Plus	1.33	0.260	1.995
S03	Tong Shan Yuan	Plus	1.35	0.240	2.025
S04	Mohamed Tarek Mohamed Khalil	Plus	1.37	0.270	2.055
S05	Rico Tan Wei Hang	Plus	1.39	0.250	2.085
S06	Marshall Chee Yu Khim	Plus	1.41	0.230	2.115
S07	Tang Kang Hao	Plus	1.43	0.260	2.145
S08	Tan Jia Xuan	Plus	1.45	0.240	2.175
S09	Teh Su Shan	Plus	1.47	0.270	2.205
S10	Yong Zi Ler	Plus	1.49	0.250	2.235
S11	Jacky Lee Qi Hong	X	1.51	0.230	2.265
S12	Esther Tong Siew Ying	X	1.53	0.260	2.295
S13	Nathaniel Isaac Raj	X	1.55	0.240	2.325
S14	Abdullah Bin Shahab	X	1.57	0.270	2.355
S15	Berlin Christopher Salomo	X	1.59	0.250	2.385
S16	Ivy Sofea Binti Mohd Azri Ngu	X	1.61	0.230	2.415
S17	Lim Ching Sze	X	1.63	0.260	2.445
S18	Eyad Mohamed Salem Maamoun Elsheikh	X	1.65	0.240	2.475
S19	Nicholas Khor Han Le	X	1.67	0.270	2.505
S20	Brandon Liu Chung Yeung	X	1.69	0.250	2.535

m0: unloaded mass including battery and equipment. l: centre-to-rotor distance. Nominal total mass includes a payload of 0.50m0, so mb = 1.50m0.

Parameter definitions and calculation

Retain the assigned unloaded mass and arm length throughout the work. Calculate derived parameters at full precision rather than copying rounded display values.

Quantity	Definition
Nominal payload and total mass	Payload = 0.50m0; mb = m0 + 0.50m0 = 1.50m0.
Increased payload and total mass	Payload = 0.80m0; m_heavy = 1.80m0.
Inertia	Use the diagonal benchmark inertia in the script below. The point payload is at the centre of mass; keep inertia unchanged during the payload test.
Rotor speed limits	0 to 800 rad/s for each rotor. Keep limits and coefficients fixed when payload changes.
Drag test	Still-air drag opposes body velocity: Fdrag = -c_drag*vb. Enable it only in the specified drag test.
Motor lag test	Time constant 0.020 s for rotor speed; initialise each rotor at hover speed. Enable lag only in the specified test.

Enter your own m0 and l from the allocation table. The example below is S01. T_rotor_max is a per-rotor limit; T_max is the total limit used in Practical 1.

```
m0 = 1.31;          % Replace with assigned unloaded mass, kg
l = 0.230;          % Replace with assigned arm length, m
g = 9.81;
mb = 1.50*m0;
m = mb;          % Actual mass used by the simulation
m_heavy = 1.80*m0;
J = diag([0.030 0.030 0.050]) ...
    * (m0/1.50) * (l/0.250)^2; % kg m^2; inputs in kg and m
Omega_max = 800;    % rad/s
kT = 2*mb*g/(4*Omega_max^2);
kMT = 0.020;        % Torque/thrust ratio, m
kM = kMT*kT;
T_hover = mb*g;
T_rotor_max = kT*Omega_max^2;
T_max = 4*T_rotor_max;
Omega_hover = sqrt(mb*g/(4*kT));
tau_m = 0.020;  % s
c_drag = 0.250;        % N s/m, drag-only test
```

For the payload-only comparison, set m = m_heavy while retaining nominal controller compensation, J, kT, kM and the original limits, as required by the coursework brief.

Rotor geometry and common conventions

Use body axes x forward, y right and z down. Rotor thrust acts along negative body z. Rotor positions below are relative to the centre of mass. View rotor spin directions from above.

Rotor	Plus position	X position	Spin
1	(l, 0, 0) front	(d, -d, 0) front left	CCW
2	(0, l, 0) right	(d, d, 0) front right	CW
3	(-l, 0, 0) rear	(-d, d, 0) rear right	CCW
4	(0, -l, 0) left	(-d, -d, 0) rear left	CW

For X geometry, d = l/sqrt(2). The arm length l is not the perpendicular lever arm about the roll or pitch axis; that lever arm is d.

Airframe reaction-torque signs about positive body z are [+1, -1, +1, -1] for rotors 1 to 4. These signs describe reaction torque, not the separate rotor gyroscopic effect.

Parameter checks

Check the units of mass, inertia, force, moment and rotor speed. With these coefficients, nominal total maximum thrust is twice nominal weight. Equal-speed hover is Omega_max/sqrt(2), approximately 565.685 rad/s. These checks do not replace validation of the nonlinear model or controller.